

Advanced .NET Code Auditing Course

.NET - Author not stated, anquanke.com.

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Content (English translation)

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Advanced .NET Code Auditing Course



Advanced .NET Code Auditing (Lesson 1): XmlSerializer Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET

The mapping rules used by XmlSerializer to convert classes are represented by metadata attributes on .NET classes. If a developer obtains external data through static methods on the Type class and calls Deserialize to deserialize XML data, this can trigger a deserialization attack. This article explains the underlying principles and code-auditing approach, with mind maps and a reproduction of the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-03-06 10:03:45; views: 794447.



Advanced .NET Code Auditing (Lesson 2): Json.Net Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET

Newtonsoft.Json makes it easy to convert between JSON and all kinds of .NET types, including objects and primitive data types. In some scenarios, using DeserializeObject with unsafe data introduces a deserialization vulnerability that enables remote code execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-03-11 10:01:22; views: 904303.



Advanced .NET Code Auditing (Lesson 3): Fastjson Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET; code security

Several deserialization vulnerabilities and bypasses affecting different versions of Java's Fastjson have been disclosed. There is also a Fastjson library for .NET. Its author describes it as the most efficient .NET component for reading and writing JSON. Its built-in JSON.ToJSON method can quickly serialize .NET objects.

[image:]Cloud Shadow Laboratory

Published: 2019-03-13 10:00:21; views: 667285.



Advanced .NET Code Auditing (Lesson 4): JavaScriptSerializer Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET

In .NET Ajax applications, the JavaScriptSerializer class commonly provides serialization functionality. In some scenarios, developers who process unsafe JSON data with Deserialize or DeserializeObject introduce a deserialization attack that enables remote code execution.

[image:]Cloud Shadow Laboratory

Published: 2019-03-20 10:06:56; views: 754267.



Advanced .NET Code Auditing (Lesson 5): .NET Remoting Deserialization Vulnerabilities

Code auditing; .NET; deserialization; code security

Security researcher @irsdI recently disclosed a potential deserialization risk in .NET Remoting applications. A deserialization vulnerability occurs when the server uses SoapServerFormatterSinkProvider as the channel sink for an HTTP channel and sets the TypeFilterLevel property for automatic deserialization to Full.

[image:]Cloud Shadow Laboratory

Published: 2019-03-26 10:06:03; views: 827004.



Advanced .NET Code Auditing (Lesson 6): DataContractSerializer Deserialization Vulnerabilities

Vulnerability analysis; .NET; deserialization

In some scenarios, using `DataContractSerializer.ReadObject` to read malicious XML data introduces a deserialization vulnerability that enables remote code execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-04-02 10:00:26; views: 621101.



Advanced .NET Code Auditing (Lesson 7): NetDataContractSerializer Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET; deserialization

Use `WriteObject` or `Serialize` to serialize an object, and `ReadObject` or `Deserialize` to deserialize an XML stream. In some scenarios, reading a malicious XML stream introduces a deserialization vulnerability that enables remote code execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-04-11 10:43:21; views: 548625.



Advanced .NET Code Auditing (Lesson 8): SoapFormatter Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET; deserialization

`SoapFormatter` and `BinaryFormatter`, which is introduced in the next lesson, are classes that implement serialization functionality within .NET. In some scenarios, processing an unsafe SOAP stream introduces a deserialization vulnerability that enables remote code

execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-04-15 10:01:23; views: 445388.



Advanced .NET Code Auditing (Lesson 9): BinaryFormatter Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET; deserialization

BinaryFormatter and SoapFormatter differ in their data-stream formats; their other functionality is similar. Deserializing an untrusted binary file can introduce a deserialization vulnerability that enables remote code execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-04-16 10:00:34; views: 456034.



Advanced .NET Code Auditing (Lesson 10): ObjectStateFormatter Deserialization Vulnerabilities

Vulnerability analysis; code auditing; .NET; deserialization

ObjectStateFormatter is generally used to serialize and deserialize state object graphs. Deserializing an untrusted binary file can introduce a deserialization vulnerability that enables remote code execution. This article explains the underlying principles and code-auditing approach and reproduces the issue.

[image:]Cloud Shadow Laboratory

Published: 2019-04-17 10:00:58; views: 598622.